



HiveBoard: An Open, Modular, 3D-Printed Benchmark of Industrial Mechanisms for Robotic and Prosthetic Manipulation

Ricardo V. Godoy¹, Enzo F. de Souza¹, Matheus P. Angarola¹, Rudy De-Xin de Lange², Juliano Negri¹, João Aires Marsicano¹, João H. Aléssio¹, Victor I. van Halst¹, Aravind Elanjimattathil Vijayan³, Gianluca Capezzuto¹, Felipe Tommaselli¹, Giuseppe Milazzo⁴, Amy M. Ramírez Sánchez⁵, Francisco Affonso⁶, Rafael R. Baptista⁷, Meiko A. van Berge⁸, Girish Chowdhary⁶, Ranulfo Bezerra⁹, Gustavo J. G. Lahr¹⁰, Lucas Ferrari Gerez², Antonio Bicchi⁴, Marcelo Becker¹

Abstract—Manipulation benchmarks commonly evaluate grasping, pick-and-place, and assembly of loose parts. Operating constrained mechanisms also requires controlled force and torque, motion along prescribed axes, and multi-step sequencing. We present  **HiveBoard**, a modular, fully 3D-printable benchmark of functional mechanisms for robotic and prosthetic manipulation. A hexagonal honeycomb base accepts 12 interchangeable attachment designs for torque, precision, and composed assembly tasks, under 13 evaluation conditions. The release includes printable geometry, URDF and USD simulation assets with Isaac Lab reinforcement-learning environments, a protocol with five trials per condition, a browser-based Evaluation Runner, and DataHive, a toolkit for collecting learning episodes. We report a cross-laboratory evaluation of five systems: a tablet-teleoperated Spot Arm, a leader-follower SO-101 arm, an ANYmal with DynaArm controlled through virtual-reality devices, a gamepad-teleoperated Franka Research 3, and a wearable Macao prosthetic hand. Each platform performed 65 trials, and four unimpaired subjects performed the same trials with their own hand. We also define the HiveScore, which weights the success rate on each condition by the completion speed relative to the human reference. None of the five platforms completed all 13 conditions, and their HiveScores range from 5.7 to 12.0 for the teleoperated arms and 41.6 for the prosthetic hand, against 85.8 to 97.8 for the human subjects. The recorded failure causes identify task-dependent limitations in grasp geometry, motion, force, and control precision. Project page: hiveboard-bench.github.io.

I. INTRODUCTION

The manipulation capabilities of robotic and prosthetic devices are benchmarked, with few exceptions, on free objects. Many objects that a deployed system must operate, however,

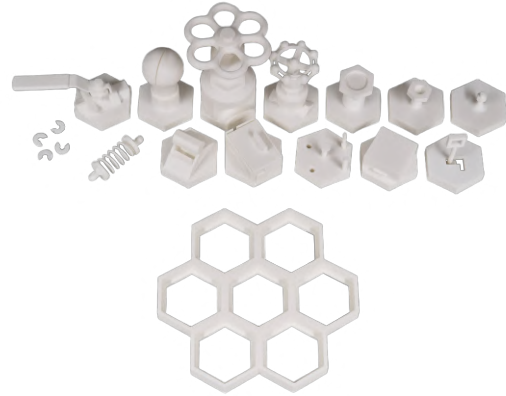


Fig. 1:  **HiveBoard**. A 3D-printed hexagonal honeycomb base accepts interchangeable attachments covering torque, precision, and composed assembly tasks. The full attachment library is reproducible on a consumer FDM printer.

are constrained mechanisms. A valve can only rotate about its stem, a screw advances along its thread against friction, a drawer slides on rails, and a lock can be opened only with a grasped, inserted, and rotated key. Operating such mechanisms depends on grasp force, kinematic reach, and sequencing in ways that grasping scores for loose objects do not capture, and such mechanisms appear in several application domains. In industry, legged platforms already inspect oil and gas installations [1] and offshore converter stations [2], yet inspection alone does not require physical interaction with the equipment, and moving from inspection to intervention means operating these same mechanisms. Prosthetic hands are increasingly designed and evaluated for activities of daily living [3], which include constrained mechanisms.

Object sets such as YCB [4], EGAD [5], and GRASPA [6], and prosthetic-hand protocols such as the AHAP [7], standardize what is grasped and how grasping is scored, but contain no mechanism to operate. Task boards built from commercial components [8], [9] capture realistic assembly and actuation, but reproducing them requires sourcing specific purchased parts. Printable benchmarks such as FurnitureBench [10] and FMB [11] remove the sourcing problem,

¹Mobile Robotics Group, University of São Paulo, Brazil. Correspondence: ricardo.godoy@usp.br.

²ARARA Lab, James Watt School of Engineering, University of Glasgow, Scotland.

³Robotics Systems Lab, ETH Zurich, Switzerland.

⁴Soft Robotics for Human Cooperation and Rehabilitation, Istituto Italiano di Tecnologia, Italy.

⁵Directorate of Technological Innovation, Universidad de las Américas Puebla, Mexico.

⁶Distributed Autonomous Systems Laboratory (DASLab), University of Illinois Urbana-Champaign, USA.

⁷Biomedical Engineering Laboratory, University of São Paulo, Brazil.

⁸Independent researcher.

⁹Graduate School of Engineering, Tohoku University, Japan.

¹⁰Instituto Israelita de Ensino e Pesquisa Albert Einstein, Hospital Israelita Albert Einstein, Brazil.

but their tasks center on part mating and insertion, without loaded mechanisms such as valves, breakers, or locks. We are not aware of a benchmark that poses functional mechanism tasks, can be printed on consumer 3D printers, and provides matching simulation assets.

This paper presents  **HiveBoard**, an open and modular manipulation benchmark (Figure 1). Its components are printable in PLA on consumer-grade FDM machines, without purchased mechanisms or metal inserts. The board is organized around a seven-cell hexagonal honeycomb base whose cells accept attachments through a shared press-fit interface, and a set of snap-on friction rings adjusts the torque of an existing attachment without printing a new one. The 12 attachment designs form 13 evaluation conditions. The torque-based tasks are a ball valve, tested with and without a friction ring, a small and a large gate valve, and a circuit breaker. The precision-based tasks are a light bulb, M8 and M30 threaded fasteners, and a threaded peg insertion. The composed assembly tasks are a push button behind a hinged cover, a lock and key, a sliding drawer, and a two-cell shock absorber, each scored stage by stage. Every attachment is also released as an articulated URDF and USD model with the joints, limits, and mass properties that common simulators require, so each task can be run physically or in simulation.

Five laboratories printed the board and ran the protocol on their own systems: a Spot Arm under tablet teleoperation, a leader-follower SO-101 arm, an ANYmal with DynaArm driven through virtual-reality devices, a gamepad-teleoperated Franka Research 3, and the Macao prosthetic hand worn by its operator. Four unimpaired subjects also completed all 13 conditions with their own hand, and their completion times are the reference against which the HiveScore normalizes platform times.

The contributions are:

- An open hardware benchmark comprising a seven-cell hexagonal base and 12 functional attachment designs evaluated in 13 conditions covering torque, precision, and composed assembly tasks, fully printable in PLA on consumer FDM printers.
- A reproducible evaluation protocol with per-attachment success criteria, timeouts, stage-wise scoring for multi-step tasks, a closed vocabulary for reporting dominant failure causes across platforms, and the HiveScore, which weights the success rate on each condition by the completion speed relative to human performance.
- Simulation assets in URDF and USD, together with a browser-based Evaluation Runner and DataHive for collecting, annotating, and validating learning episodes.
- A cross-laboratory evaluation of five manipulation systems and a four-subject human reference, reporting success, completion time, stage completion, and failure causes for every condition.

II. RELATED WORK

A. Object sets and grasping benchmarks

The YCB object and model set [4] established a shared set of physical objects with mesh models and benchmarking

protocols for manipulation research. EGAD [5] contributed a printable set of procedurally generated objects with controlled geometric diversity, and GRASPA [6] defined printable layouts and scoring to benchmark grasp pipelines on physical arms while accounting for the limits of the test platform. These efforts standardize the objects and, in part, the procedure, but they do not define success criteria for constrained mechanisms, such as actuating a valve against resistance or threading a fastener.

B. Task boards for industrial manipulation

Closer to our setting, the NIST assembly task boards [8] arrange commercial connectors, fasteners, and belt drives on standardized plates, with disassembly and assembly test modes and scoring that has anchored several robot competitions. IndustReal [12] developed simulation environments and learned policies around the same component set, transferring contact-rich assembly skills from simulation to reality. The Robothon Grand Challenge and the euROBIN skill versatility challenge evaluate manipulation on an electronic task board whose embedded microcontroller verifies outcomes and logs trials [9]. BusyBox [13] provides six reconfigurable modules for buttons, switches, sliders, wires, a dial, and a display, with printable components, optional electronic state logging, and manipulation demonstrations.

C. Benchmarking prosthetic and anthropomorphic hands

The Anthropomorphic Hand Assessment Protocol (AHAP) scores a hand across 26 postures and tasks built on 25 YCB objects, and its grasping ability score has been used to compare robotic and prosthetic hand designs [7]. The Southampton Hand Assessment Procedure (SHAP) instead times the hand on 26 tasks, including abstract grasps and activities of daily living such as turning a key and a door handle, and scores their completion times relative to normative data from unimpaired subjects [14]. The  **HiveBoard** protocol defines success by mechanism state and prescribes no gripper or controller, so a prosthetic hand and a robot gripper are evaluated on the same tasks.

D. Printable manipulation benchmarks

FurnitureBench [10] provides printable furniture, standardized initialization, demonstration data, and FurnitureSim for benchmarking long-horizon furniture assembly. FMB [11] provides printable objects, assembly boards, demonstrations, and policies for grasping, reorientation, and multi-stage insertion. Table I compares the physical benchmarks most closely related to  **HiveBoard**, whose distinction is the combination of fully printable functional mechanisms, interchangeable mounting, adjustable ball-valve resistance, and a protocol applied to both robot-mounted grippers and a wearable prosthetic hand.

III. THE **HIVEBOARD** BENCHMARK

A. Hexagonal base

The base consists of seven hexagonal cells in a hive configuration, one central cell surrounded by six outer cells

TABLE I: **Feature comparison of physical manipulation benchmarks.** Checkmarks indicate features documented in the cited works, while dashes indicate features not established by those sources. The criteria distinguish task hardware, manipulation operations, released resources, and reported experiments.

	Criterion	AHAP [7]	NIST [8] boards	DR.J [9]	Furniture- Bench[10]	FMB [11]	BusyBox [13]	HiveBoard (ours)
Hardware	Printable task parts	–	–	–	✓	✓	✓	✓
	Interchangeable task modules	–	–	–	–	–	✓	✓
	Embedded state sensing	–	–	✓	–	–	✓	–
Operations	Insertion / assembly	–	✓	✓	✓	✓	✓	✓
	Threaded fastening	–	✓	–	✓	–	–	✓
	Valve actuation	–	–	–	–	–	–	✓
Resources	Simulation support	–	–	–	Env.	–	–	Mod. + Env.
	Learning dataset	–	–	–	✓	✓	✓	✓
	Collection software	–	–	–	✓	✓	✓	✓
Experiments	Learned-policy results	–	–	–	✓	✓	✓	✓
	Prosthetic-hand results	✓	–	–	–	–	–	✓

Definitions. Printable parts are manipulated objects or mechanism components. A checkmark does not imply that all hardware is printable. Interchangeable modules share a mechanical interface for rearranging task mechanisms. Embedded sensing refers to task hardware and is optional in BusyBox. Insertion includes keys and connectors. Simulation: *Env.* denotes a provided simulation environment; *Mod.* denotes URDF/USD assets. A learning dataset contains demonstration trajectories for policy training. Experiments refer to reported evaluations, not potential applications.

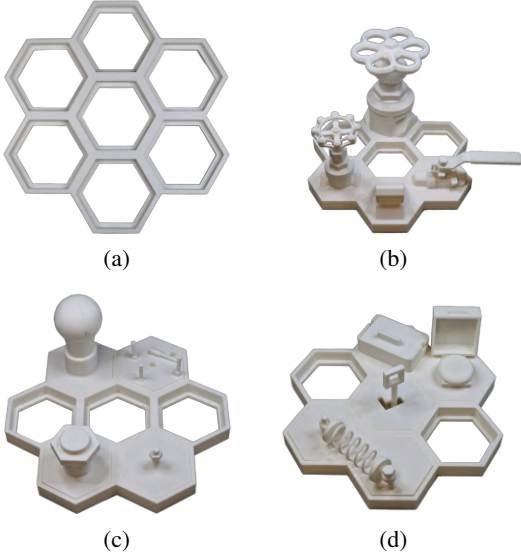


Fig. 2: (a) The  **HiveBoard base, a seven-cell hexagonal honeycomb**. The base can be arranged with different attachments to evaluate skill sets: (b) **torque-based tasks**, (c) **precision-based tasks**, and (d) **composed assembly tasks**.

(Figure 2a). Each cell is an open frame that accepts the hexagonal footprint of an attachment, with mounting geometry shared across attachment designs. Base units can be combined when an attachment occupies more than one cell or when a larger evaluation scene is needed. Attachments are held by a press fit without fasteners and can be exchanged by hand.

B. Attachment library

The library contains 12 functional attachment designs in three categories, shown in Figure 3. The ball valve can

be evaluated both with and without a friction ring, which increases the torque required to complete the task. Therefore, the benchmark comprises 13 evaluation conditions.

1) *Torque-based tasks* (Figure 2b): This category comprises a ball valve, two gate valves whose handle geometries and sizes require different torques, and a single-pole circuit breaker. The ball valve is paired with four snap-on friction rings that fit over the valve body, each setting a different rotational friction level, so the same physical valve can be tested at multiple torque settings.

2) *Precision-based tasks* (Figure 2c): This category comprises a printed light bulb and socket, M8 and M30 threaded fasteners, and a peg-insertion plate. The two fasteners differ in diameter and pitch and therefore in the grasp and alignment they require. The peg-insertion plate has four identical threaded sockets, with three matching pegs already seated and one peg free. The pegs are 8 mm threaded pins, so the operator grasps the free peg, aligns it with the empty socket, and then rotates it down until it seats.

3) *Composed assembly tasks* (Figure 2d): This category comprises a push button hidden by a hinged cover, a key-and-lock mechanism that requires grasp, vertical insertion, and rotation, a sliding drawer with a recessed handle, and a shock-absorber model that occupies two adjacent cells, in which two hexagonal end pieces are joined by a coil spring with a pin running along the spring axis. These attachments require multiple sub-actions and are scored stage by stage.

C. Manufacturing

All parts print in PLA on consumer-grade FDM machines with a 0.4 mm nozzle, 0.20 mm layer height, four perimeter walls, five top and bottom layers, 50 mm/s print speed, nozzle temperature between 200 and 220°C, and bed temperature between 50 and 60°C. Cooling is set to 100% and a skirt or brim is used for adhesion, with supports only where required.

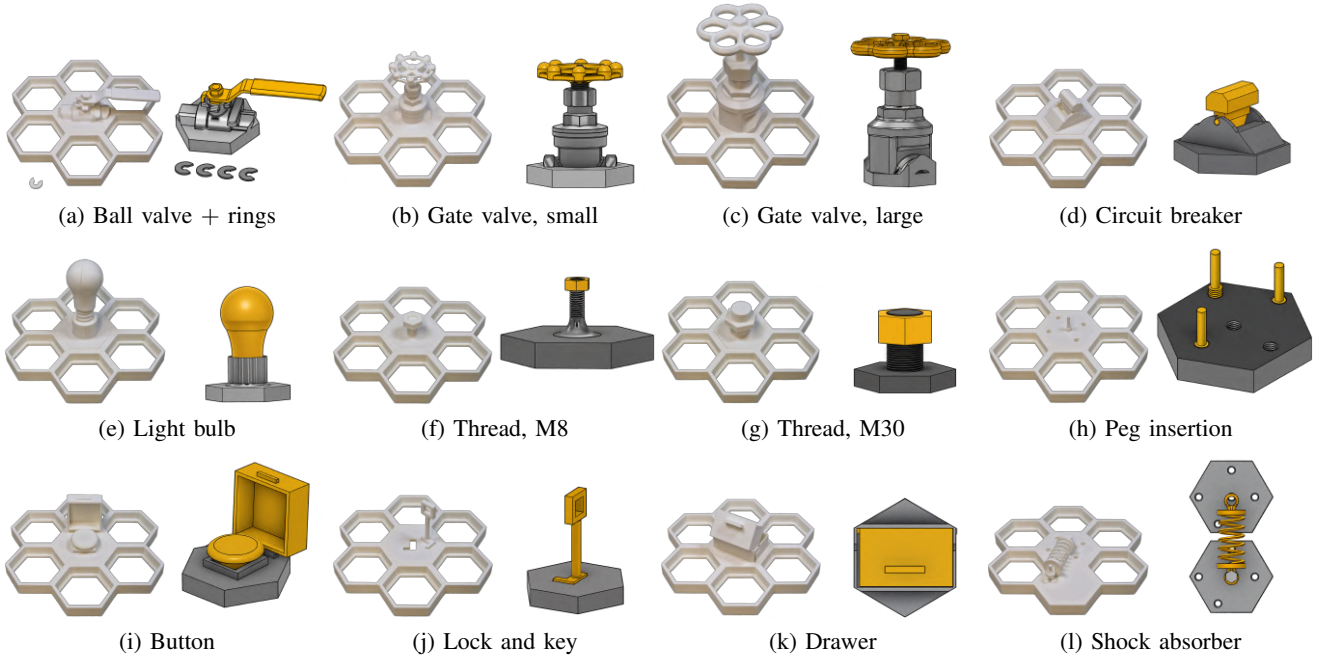


Fig. 3: **The  HiveBoard attachment library.** Each pair of images shows the printed part (left) next to its digital twin (right). **Torque-based attachments** (a)–(d), with the friction ring set in (a) modulating the breakaway torque of the ball valve. **Precision-based attachments** (e)–(h). **Composed assembly attachments** (i)–(l), where (l) occupies two adjacent cells. The moving parts of each object are shown in yellow in the digital twin.

TABLE II: **Infill and recommended print orientation for each class of printed part.**

Part class	Infill	Orientation
Base structure	20%	flat
Covers	15%	flat
Torque components	30%	handle up
Threads, screws, pegs	25%	vertical
Nuts	25%	flat
Drawer	25%	flat
Shock absorber parts	25%	sideways

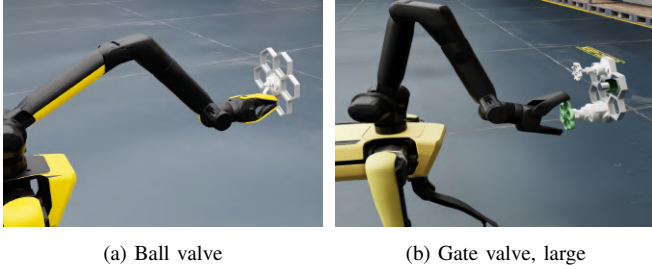


Fig. 4: **Simulation validation with Spot.** Two torque-based tasks are shown: (a) **ball valve** and (b) **large gate valve**.

A build volume of 300×300 mm allows the base to be printed as a single piece, improving dimensional accuracy and press-fit consistency across printers and laboratories. Infill varies by part type to balance mass, strength, and print time, and parts are oriented to align load paths with the layer planes wherever possible (Table II).

D. Simulation assets

Every attachment and the base are released as simulation assets, organized per assembly with URDF descriptions, USD exports, and OBJ visual and collision meshes. The ball valve and the circuit breaker expose revolute joints with their physical angular ranges. The gate valves, the two threaded fasteners, and the light bulb socket model screw motion as a pair of coupled joints, a continuous rotation about the thread axis and a prismatic translation along it, which reproduces the advance of the part without a native helical joint type. The button combines a revolute joint for the hinged cover with a prismatic joint for the press, and the drawer, the lock, and the shock absorber carry their articulation in the USD assets. Mass properties are estimated from the volume of each part and the density of PLA.

We verified that the assets load in MuJoCo, PyBullet, and Isaac Sim, that each joint articulates through its physical range, and that the collision geometry does not self-intersect in the rest configuration. Joint friction, mass, and inertia are nominal values, because the contact properties of printed parts depend on the printer, filament, and slicer calibration at each laboratory, and users can override them after identifying the parameters on their own print. Figure 4 shows two of the validated assets.

E. Reinforcement-learning environments

We provide Isaac Lab environments [15] for reinforcement-learning (RL) research, enabling users to evaluate  **HiveBoard** without physical hardware and develop policies for sim-to-real transfer. Each task–platform

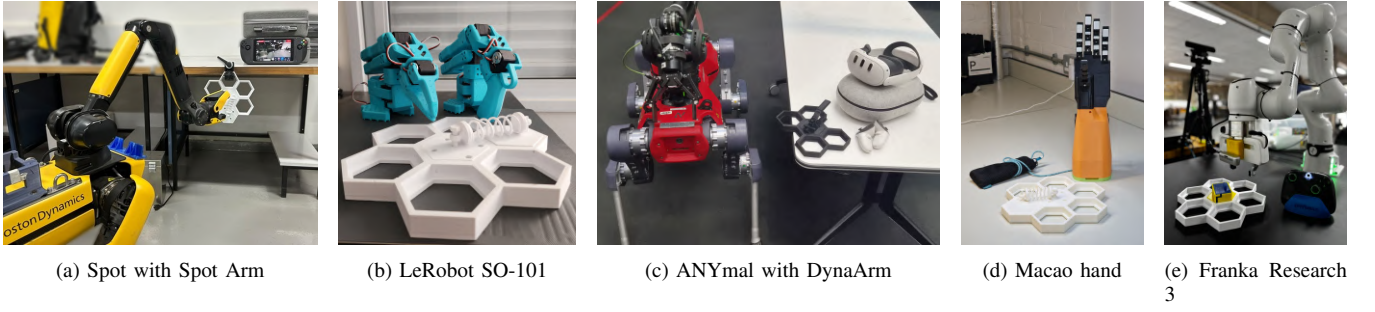


Fig. 5: The five manipulation systems, photographed at their laboratories: (a) Spot with Spot Arm, (b) LeRobot SO-101, (c) ANYmal with DynaArm, (d) Macao hand, and (e) Franka Research 3. Each laboratory printed its own board from the released geometry.

combination defines a separate environment, while multiple instances of each scene can be simulated in parallel for policy optimization. This organization allows task configurations, robot models, controllers, observations, and termination conditions to be modified independently.

The RL objective is to learn a policy $\pi(a_t | o_t)$ that maps observation o_t to action a_t while maximizing the expected discounted return

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(o_t, a_t) \right], \quad (1)$$

where τ denotes an on-policy trajectory, $\gamma \in [0, 1)$ is the discount factor, and $r(o_t, a_t)$ is the reward obtained at time step t . The environments are compatible with RSL-RL [16], which supports proximal policy optimization (PPO) [17] and teacher-student distillation [18].

We first define the base observation space as

$$o_t = \left[x_t^{ee,b}, x_t^{obj,b}, a_{t-1}, z_t^{priv} \right], \quad (2)$$

where $x_t^{ee,b}$ and $x_t^{obj,b}$ denote the current end-effector and object poses, respectively, expressed in the robot-base frame. Each pose comprises a three-dimensional position and an orientation quaternion. The term a_{t-1} represents the previous action, while z_t^{priv} groups task-dependent privileged information obtained from the simulator (e.g., the direction of rotation of a valve). This common representation allows each environment to include the state variables required by its mechanism. The observation space can also be extended with exteroceptive measurements to support policies intended for deployment without access to privileged simulator information.

The action is the commanded end-effector state, $a_t = x_t^{cmd,b}$, which combines the commanded end-effector pose increments $(\Delta p_t, \Delta \phi_t)$ with the gripper-opening command g_t . A platform-specific controller maps the commanded state to joint-position targets, providing a common policy interface across manipulators.

The released environments were tested using operator-issued actions to validate the simulated interactions. The current release does not yet include demonstration datasets, task-specific reward functions, or trained policies. Instead, it

provides a common simulation foundation on which these components can be developed and evaluated.

IV. EVALUATION PROTOCOL

A. Platforms

Figure 5 shows the five systems. Platform A is a Boston Dynamics Spot with the Spot Arm under tablet teleoperation. Platform B is a leader-follower LeRobot SO-101 arm [19], [20]. Platform C is an ANYmal [21] with a DynaArm and a Robotiq 2F-140 gripper, operated through a Meta Quest 3 headset and hand controllers. Platform D is the Macao prosthetic hand [22], worn by an able-bodied operator who selects grasps with analog buttons. Platform E is a Franka Research 3 [23] with the standard Franka Hand, teleoperated with a gamepad. Platforms A and C used vertical boards, and Platforms B, D, and E horizontal ones.

B. Protocol

The evaluation comprises five recorded trials for each of the 13 conditions, giving 65 trials per platform. The protocol defines success by the operation completed on the attachment within the specified timeout (Table III). The experiments in this paper use operator-issued task commands, and Platform C additionally uses a learned whole-body controller [24] to track them.

Before recording, operators perform familiarization trials for each condition until three consecutive completion times fall within 20% of each other, and practice trials are excluded from the scored evaluation. The board orientation, attachment configuration, control mode, and controller parameters are held fixed within each recorded condition. Each trial starts from the specified initial mechanism state and a fixed neutral end-effector pose. Timing begins with the first commanded task motion and stops at success, an unsuccessful termination, a safety stop, or the condition timeout, and only successful trials receive a completion time. Trials involving damage or displacement remain in the log, with the event described in the notes.

C. Metrics

For each platform and condition, we report the number of successful trials out of five and the median completion

TABLE III: Task specifications for all 13 evaluation conditions. Stages apply to composed assembly tasks. Record the last completed stage, with 0 indicating that no stage was completed.

Category	Condition	Success criterion	Timeout (s)	Stages
Torque	Ball valve	Handle rotated 90° from closed to open	60	Single action
	Ball valve + ring	Handle rotated 90° from closed to open, ring fitted	90	Single action
	Gate valve (small)	Stem rotated one full turn from the closed position	90	Single action
	Gate valve (large)	Stem rotated one full turn from the closed position	120	Single action
	Circuit breaker	Toggle moved from one state to the other and held	60	Single action
Precision	Light bulb	Bulb fully threaded until seated	120	Single action
	Thread (M8)	Bolt fully threaded along the available length	120	Single action
	Thread (M30)	Bolt fully threaded along the available length	120	Single action
	Peg insertion	Free peg threaded into the empty socket until it seats	120	Single action
Composed assembly	Button	All stages completed within the timeout	60	(1) Open the cover → (2) press the button until actuation
	Lock and key	All stages completed within the timeout	180	(1) Grasp the key → (2) insert vertically → (3) rotate to unlock
	Drawer	All stages completed within the timeout	120	(1) Grasp the handle → (2) pull open → (3) push closed
	Shock absorber	All stages completed within the timeout	180	(1) Grasp the pin → (2) align with the hole → (3) insert fully

time of the successful trials. Completion times differ widely between conditions, from a single lever push to a multi-stage assembly, so we also express them relative to unimpaired human performance, as timed hand-function tests do when they score task times against normative data from unimpaired subjects [14], [25]. The reference time h_a of condition a is the median completion time of unimpaired subjects performing it with their own hand, and the slowdown of a platform on that condition is its median time divided by h_a .

To combine success and speed, platform p scores, on each condition, its success rate multiplied by the ratio of the reference time to its median time $t_{p,a}$. That ratio is capped at one, so finishing one condition faster than the human reference cannot offset a failure on another, and a condition with no successful trial scores zero. With $s_{p,a}$ successful trials out of five, the HiveScore is the mean over the 13 conditions, expressed out of 100,

$$HS_p = \frac{100}{13} \sum_{a=1}^{13} \frac{s_{p,a}}{5} \min\left(1, \frac{h_a}{t_{p,a}}\right), \quad (3)$$

and it reaches 100 only when every trial succeeds within the human reference time. Since platforms with the same HiveScore can differ in how many conditions they complete and in how fast they complete them, we report it together with the success count and the slowdown.

D. Trial logging and learning data

The release includes the protocol, task instructions, and trial templates in CSV and spreadsheet formats, and a browser-based Evaluation Runner that displays the task criteria, times each trial, and exports the records. A complete submission requires all 65 trial entries, a platform description, a setup photograph, and an external-camera recording of each trial. DataHive, a separate toolkit adapted from the Oopsie Data workflows, lets laboratories associate HDF5 state and command records and MP4 camera recordings with

task annotations, validate the episodes, and upload them to a private Hugging Face dataset repository. Using demonstrations collected with DataHive, we have also fine-tuned and evaluated the vision-language-action model $\pi_{0.5}$ [26] on  **HiveBoard** tasks. These learned-policy results, along with the dataset used to fine-tune them, will be reported on the project page and in the full version of this paper.

V. EXPERIMENTAL VALIDATION

Each laboratory printed the board locally from the released STL files with the recommended print profile and ran the protocol of Section IV on its own hardware. All five platforms completed 65 recorded trials, and four unimpaired subjects completed the same protocol to provide the reference times.

A. Human reference

Four unimpaired subjects performed all 13 conditions with their own right hand, with five recorded trials per condition after familiarization and the same timing rules as the platforms, on a board mounted horizontally on a desk in front of them. The subjects were right-handed and between 20 and 30 years old. The study was approved by the ethics committee of the University of Glasgow, and the subjects gave informed consent. All 260 human trials succeeded, each at the first attempt and without a regasp. Table IV lists the reference time h_a for each condition.

B. Validation results

Table V reports the successful trials and median completion times. All five platforms completed four conditions in every trial (both ball-valve configurations, the large gate valve, and the button). The circuit breaker and the lock were completed by all five, with a single failure on Platform B and on Platform E respectively, and the light bulb by every platform except Platform E. The M8 fastener and the peg, the two parts that require a small threaded item to be held while it is turned, failed on all five. The small gate valve was

TABLE IV: **Human reference completion times.** h_a is the median over the four subjects of each subject’s median over five trials, and the range gives the lowest and highest subject medians. All 260 trials succeeded.

Condition	h_a (s)	Subject medians (s)
Ball valve	1.27	1.20 to 1.81
Ball valve + ring	1.54	1.31 to 1.96
Gate valve (small)	2.29	1.72 to 2.83
Gate valve (large)	2.62	1.61 to 3.83
Circuit breaker	1.45	1.12 to 1.47
Light bulb	4.29	4.09 to 4.62
Thread (M8)	5.91	4.79 to 9.05
Thread (M30)	11.46	7.77 to 13.41
Peg insertion	8.18	7.54 to 8.96
Button	2.43	1.89 to 2.73
Lock and key	3.16	2.95 to 3.45
Drawer	4.06	3.34 to 5.13
Shock absorber	2.68	2.43 to 3.87

TABLE V: **Successful trials out of five and median completion time of the successful trials (s) for each platform and condition.** “–” marks a median that is undefined because no trial succeeded.

Condition	Successes					Median time (s)				
	A	B	C	D	E	A	B	C	D	E
Ball valve	5	5	5	5	5	10.5	9.0	7.8	2.1	14.1
Ball valve + ring	5	5	5	5	5	16.9	9.0	9.2	2.2	14.7
Gate valve (small)	5	0	5	5	5	19.9	–	23.1	2.7	34.2
Gate valve (large)	5	5	5	5	5	17.4	10.0	16.8	2.3	21.8
Circuit breaker	5	4	5	5	5	15.6	8.5	6.7	1.6	13.8
Light bulb	5	5	5	5	0	27.3	83.0	19.3	52.1	–
Thread (M8)	0	0	0	0	0	–	–	–	–	–
Thread (M30)	0	0	0	4	0	–	–	–	102.6	–
Peg insertion	0	0	0	0	0	–	–	–	–	–
Button	5	5	5	5	5	23.9	10.0	13.3	3.0	23.1
Lock and key	5	5	5	5	4	40.4	15.0	21.4	15.1	60.2
Drawer	0	5	0	5	1	–	23.0	–	39.9	57.1
Shock absorber	0	5	0	4	5	–	16.0	–	39.4	29.2

completed by every platform except Platform B, the M30 fastener only by Platform D, the drawer by Platforms B and D and once by Platform E, and the shock absorber by Platforms B, D, and E. The friction ring added 6.4 s to the median ball-valve time on Platform A, 1.4 s on Platform C, 0.6 s on Platform E, and 0.1 s on Platform D, while Platform B recorded 9.0 s in both configurations.

Table VI lists the recorded failure causes, which differ by platform. On Platform A, the wrist does not rotate cleanly about a fixed axis under tablet teleoperation, so both threaded fasteners and the peg fail as kinematic limits, and the parallel jaw could not hold the recessed drawer handle or the shock-absorber pin, which are recorded as grasp geometry. Platform B records 20 of its 21 failures as kinematic limits, and Platform E records kinematic limits on all four threaded conditions, including the light bulb, because none of the threaded parts could be aligned with its mate. Platform C records control precision on all 25 failures, which the operator attributes to alignment through the virtual-reality interface. On Platform D, the M8 screw

TABLE VI: **Dominant failure causes of the unsuccessful trials, with trial counts out of five.** K: kinematic limit; G: grasp geometry; C: control precision; F: force limit; S: slip; O: other. Conditions without failures are omitted.

Condition	A	B	C	D	E
Gate valve (small)	–	K5	–	–	–
Circuit breaker	–	O1	–	–	–
Light bulb	–	–	–	–	K5
Thread (M8)	K5	K5	C5	F4, O1	K5
Thread (M30)	K5	K5	C5	F1	K5
Peg insertion	K5	K5	C5	C5	K5
Lock and key	–	–	–	–	S1
Drawer	G5	–	C5	–	G4
Shock absorber	G5	–	C5	F1	–

TABLE VII: **Platform summary.** Completed: conditions with at least one success. Slowdown: median $t_{p,a}/h_a$ over completed conditions. Human scores are computed for each subject against the other three.

Platform	Successes	Completed	Slowdown	HiveScore
A	40/65	8	9.3	7.0
B	44/65	9	5.8	12.0
C	40/65	8	6.0	10.4
D	53/65	11	1.6	41.6
E	40/65	9	10.9	5.7
Human subjects	260/260	13	–	85.8 to 97.8

head could not be held against thread friction, and eleven of its twelve failures are force limits or control precision. Stage-wise scoring separates failures with the same outcome, since the drawer and the shock absorber failed before stage 1 on Platform A, while Platform C grasped both parts in all five trials and then failed at the subsequent motion or alignment.

C. HiveScore

Table VII summarizes the human-referenced scores. Scored against the other three, the four human subjects reach 85.8 to 97.8. The Macao reaches 41.6 and the four teleoperated arms 5.7 to 12.0. On the conditions they completed, the arms took a median of 5.8 to 10.9 times the reference time, against 1.6 times for the Macao, and if every successful trial had matched the reference time, each score would equal the success rate, 62 to 68 for the arms and 82 for the Macao. Platforms A, C, and E succeeded in the same 40 trials, and their scores differ through speed alone.

D. Limitations

Each platform was operated by a single operator at one laboratory, so operator skill and platform capability are not separable in these data. The human reference comes from four right-handed subjects at one laboratory. This version of the paper also doesn’t systematically address sim2real deployments and evaluation of learning-based methods.

VI. CONCLUSION

We present  **HiveBoard**, an open, modular benchmark of functional mechanisms modeled on industrial fittings

and printed in PLA on consumer FDM machines, released under a permissive license with URDF and USD assets, Isaac Lab RL environments, the evaluation protocol, and logging templates. Five systems completed the protocol at 65 trials each, and none completed the full library, with the Spot Arm and the ANYmal succeeding on eight of the 13 conditions, the SO-101 and the Franka on nine, and the Macao on eleven. The recorded failure causes differ by platform, with kinematic limits dominating on the Spot Arm, the SO-101, and the Franka, control precision on every failed ANYmal trial, and force limits and control precision on the Macao. Against the times of four unimpaired subjects, the teleoperated arms score 5.7 to 12.0 on the HiveScore and the Macao 41.6, compared with 85.8 to 97.8 for the subjects. The same tasks are available in simulation, where the Isaac Lab environments share one observation and action interface across manipulators, and demonstrations collected with DataHive were used to fine-tune the VLA model $\pi_{0.5}$, whose results are reported in the full version of this paper.

ACKNOWLEDGMENTS

This work was supported by FAPESP, Grants #2025/08520-0, #2025/22381-3, #2025/20858-7, and #2025/04308-7; by Petrobras in partnership with the USP and FAFQ, under Cooperation Agreements #2023/00016-6 and #2023/00013-7; by ANYbotics AG; by Royal Society Research Grant #251405.

REFERENCES

- [1] C. D. Bellicoso, M. Bjelonic, L. Wellhausen, D. Sako, K. Holtmann, F. Günther, M. Tranzatto, P. Fankhauser, and M. Hutter, “Advances in real-world applications for legged robots,” *Journal of Field Robotics*, vol. 35, no. 8, pp. 1311–1326, 2018.
- [2] C. Gehring, P. Fankhauser, L. Isler, R. Diethelm, S. Bachmann, M. Potz, L. Gerstenberg, and M. Hutter, “ANYmal in the field: Solving industrial inspection of an offshore HVDC platform with a quadrupedal robot,” in *Field and Service Robotics: Results of the 12th International Conference*. Springer, 2021, pp. 247–260.
- [3] H. Yang, Z. Tao, J. Yang, W. Ma, H. Zhang, M. Xu, M. Wu, S. Sun, H. Jin, W. Li *et al.*, “A lightweight prosthetic hand with 19-DOF dexterity and human-level functions,” *Nature Communications*, vol. 16, no. 1, p. 955, 2025.
- [4] B. Calli, A. Walsman, A. Singh, S. Srinivasa, P. Abbeel, and A. M. Dollar, “Benchmarking in manipulation research: Using the Yale-CMU-Berkeley object and model set,” *IEEE Robotics & Automation Magazine*, vol. 22, no. 3, pp. 36–52, 2015.
- [5] D. Morrison, P. Corke, and J. Leitner, “EGAD! an evolved grasping analysis dataset for diversity and reproducibility in robotic manipulation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 4368–4375, 2020.
- [6] F. Bottarel, G. Vezzani, U. Pattacini, and L. Natale, “GRASPA 1.0: GRASPA is a robot arm grasping performance benchmark,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 836–843, 2020.
- [7] I. Llop-Harillo, A. Pérez-González, J. Starke, and T. Asfour, “The anthropomorphic hand assessment protocol (AHAP),” *Robotics and Autonomous Systems*, vol. 121, p. 103259, 2019.
- [8] K. Kimble, K. Van Wyk, J. Falco, E. Messina, Y. Sun, M. Shibata, W. Uemura, and Y. Yokokohji, “Benchmarking protocols for evaluating small parts robotic assembly systems,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 883–889, 2020.
- [9] P. So, A. Sarabakha, F. Wu, U. Culha, F. J. Abu-Dakka, and S. Haddadin, “Digital robot judge: Building a task-centric performance database of real-world manipulation with electronic task boards,” *IEEE Robotics & Automation Magazine*, vol. 31, no. 4, pp. 32–44, 2024.
- [10] M. Heo, Y. Lee, D. Lee, and J. J. Lim, “FurnitureBench: Reproducible real-world benchmark for long-horizon complex manipulation,” *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1863–1891, 2025.
- [11] J. Luo, C. Xu, F. Liu, L. Tan, Z. Lin, J. Wu, P. Abbeel, and S. Levine, “FMB: a functional manipulation benchmark for generalizable robotic learning,” *The International Journal of Robotics Research*, vol. 44, no. 4, pp. 592–606, 2025.
- [12] B. Tang, M. A. Lin, I. Akinola, A. Handa, G. S. Sukhatme, F. Ramos, D. Fox, and Y. Narang, “IndustReal: Transferring contact-rich assembly tasks from simulation to reality,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [13] D. Fortier, T. Adamson, T. Hellebrekers, T. LaScala, K. Ennin, M. Murray, A. Kolobov, and G. Mullins, “Benchmarking affordance generalization with BusyBox,” *arXiv preprint arXiv:2602.05441*, 2026.
- [14] C. M. Light, P. H. Chappell, and P. J. Kyberd, “Establishing a standardized clinical assessment tool of pathologic and prosthetic hand function: Normative data, reliability, and validity,” *Archives of Physical Medicine and Rehabilitation*, vol. 83, no. 6, pp. 776–783, 2002.
- [15] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Munoz, X. Yao, R. Zurbrugg, N. Rudin *et al.*, “Isaac Lab: A GPU-accelerated simulation framework for multi-modal robot learning,” *arXiv preprint arXiv:2511.04831*, 2025.
- [16] C. Schwarke, M. Mittal, N. Rudin, D. Hoeller, and M. Hutter, “RSL-RL: A learning library for robotics research,” *arXiv preprint arXiv:2509.10771*, 2025.
- [17] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [18] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, vol. 15. PMLR, 2011, pp. 627–635.
- [19] R. Cadene, S. Alibert, F. Capuano, M. Aractingi, A. Zouitine, P. Kooijmans, J. Choghari, M. Russi, C. Pascal, S. Palma *et al.*, “LeRobot: An open-source library for end-to-end robot learning,” in *International Conference on Learning Representations*, 2026.
- [20] R. Knight, P. Kooijmans, R. Cadene, S. Alibert, M. Aractingi, D. Aubakirova, A. Zouitine, R. Martino, S. Palma, C. Pascal, and T. Wolf, “Standard Open SO-100 & SO-101 Arms,” <https://github.com/TheRobotStudio/SO-ARM100>, 2024.
- [21] M. Hutter, C. Gehring, D. Jud, A. Lauber, C. D. Bellicoso, V. Tsounis, J. Hwangbo, K. Bodie, P. Fankhauser, M. Bloesch, R. Diethelm, S. Bachmann, A. Melzer, and M. Hoepflinger, “ANYmal - a highly mobile and dynamic quadrupedal robot,” in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 38–44.
- [22] R. D.-X. de Lange, A. Japes, and L. Ferrari Gerez, “Open-source prosthetic hands for STEM education and community use,” in *Proceedings of the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE, Jul. 2026.
- [23] S. Haddadin, S. Parusel, L. Johannsmeier, S. Golz, S. Gabl, F. Walch, M. Sabaghian, C. Jähne, L. Hausperger, and S. Haddadin, “The Franka Emika robot: A reference platform for robotics research and education,” *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, 2022.
- [24] A. E. Vijayan, A. Cramariuc, M. Risiglione, C. Gehring, and M. Hutter, “Multi-critic learning for whole-body end-effector twist tracking,” in *Proceedings of The 9th Conference on Robot Learning*. PMLR, 2025, pp. 1470–1485.
- [25] J. G. M. Burgerhof, E. Vasluian, P. U. Dijkstra, R. M. Bongers, and C. K. van der Sluis, “The Southampton Hand Assessment Procedure revisited: A transparent linear scoring system, applied to data of experienced prosthetic users,” *Journal of Hand Therapy*, vol. 30, no. 1, 2017.
- [26] Physical Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai *et al.*, “ $\pi_{0.5}$: a vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025.